

Comprehensively Understanding Remote Sensing Scene Segmentation via VFM and MLLM Collaboration

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Abstract—Single modality data have inherent limitations for remote sensing (RS) scene segmentation, and model performance is often constrained by the reliance on large-scale manually annotated pixel-level labels. Vision Foundation Model (VFM) and Multimodal Large Language Model (MLLM) demonstrate remarkable capabilities in visual perception and textual understanding, respectively. In this work, we fully leverage a VFM to extract visual feature representations and employ a MLLM to understand RS-Scenes and generate RS captions that describe scene elements.

We propose VMCSeg, an RS-Scenes segmentation framework in which a VFM collaborates with an MLLM. Specifically, MLLM are leveraged to generate high-quality captions describing dense RS-Scenes, and then CLIP extracts effective semantic information. Meanwhile, DINOv3 is used to extract dense land-cover features, and the Bridge Adapter module is designed to enable interaction between general and detailed features. Furthermore, a Vision Language CrossFormer module facilitates interaction between textual and visual features for scene segmentation. The MLLM is instantiated with ChatGPT-4o. Our method achieves excellent performance on publicly available semantic segmentation datasets.

Index Terms—remote sensing, scene segmentation, Vision Foundation Model, Multimodal Large Language Model.

I. INTRODUCTION

Semantic segmentation [1] is a fundamental task in remote sensing interpretation, with the goal of assigning each pixel to a specific land-cover category. However, its performance is often constrained when dealing with high-resolution, complex, and densely overlapping remote sensing (RS) scenes [2], [3]. Traditional methods primarily rely on single-modality data [4], which limits the ability to capture detailed land-cover information and frequently results in misclassifications and ambiguous boundaries.

As such, perceiving and segmenting complex, overlapping remote sensing scenes using multiple modalities is crucial, as multimodal data more accurately reflect real-world sensing. In this work, we leverage Multimodal Large Language Model [5] to generate descriptive captions that guide the fusion of visual features for remote sensing land-cover segmentation.

Researchers have explored generating textual descriptions from RS-Scenes to facilitate the fused perception and segmen-

tation of textual and visual information [6], [7]. For complex and densely packed scenes, accurately understanding scene element information is even more critical.

To fully explore the potential of large models in remote sensing (RS) tasks, this work proposes a semantic segmentation method in which a Vision Foundation Model and a Multimodal Large Language Model collaborate. Unlike simple single modality image approaches or basic text generation methods, this work leverages an LLM to generate high quality captions that describe and capture RS-Scenes. Dense visual features are extracted via Vision Foundation Model, and the two modalities are then fused for RS-Scenes segmentation. The contributions of this work as follows:

- DINOv3 is used to extract dense remote sensing features, while a lightweight encoder and bridging adapter capture detailed and specific information.
- We leverage an MLLM to perceive remote sensing scenes and generate corresponding descriptive text.
- A Vision-Language cross-attention module is designed to fuse textual semantics with visual representations.

II. RELATED WORK

A. Remote Sensing Scene Semantic Segmentation

Supervised deep learning has significantly advanced semantic segmentation in remote sensing Scenes. For example, multi-scale convolutional neural networks (CNNs) have been proposed to efficiently extract land-cover features [8], [9], while various attention mechanisms [10], [11] have been designed to address the unique characteristics of remote sensing imagery. Additionally, some studies have developed hybrid architectures that combine Transformers and CNNs to achieve more accurate land-cover segmentation, exploring the potential of Transformer backbones for remote sensing tasks [12]–[14]. However, these approaches are limited to single modality data, which greatly restricts segmentation performance.

B. Vision-Language Research in Remote Sensing Scene

Due to the limitations of single modality methods, recent studies have explored the use of multimodal data [15], [16]

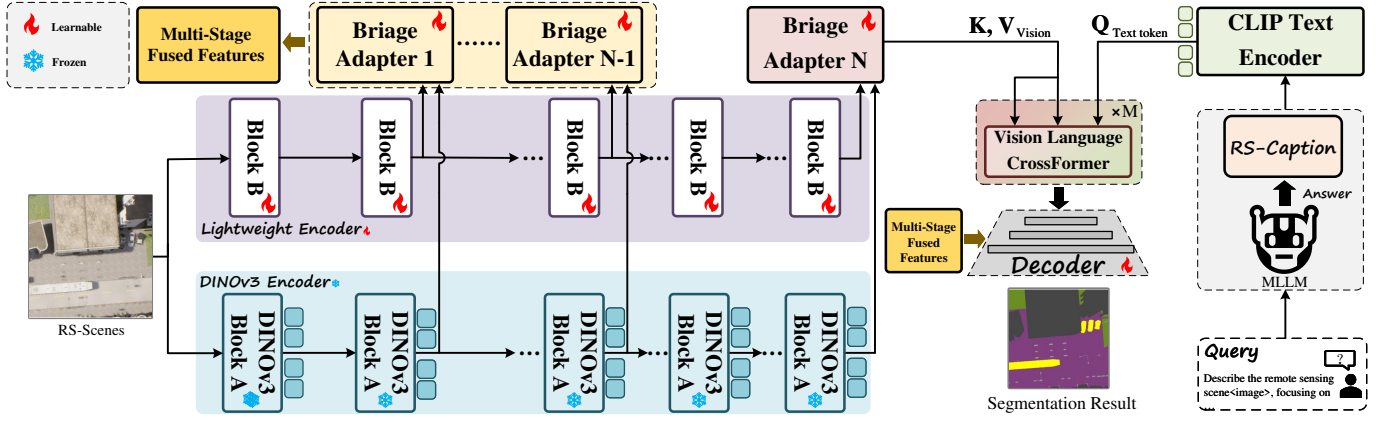


Fig. 1. The overall framework of VMCSEg. The MLLM perceives RS scenes and generates captions for textual semantics; the vision encoder extracts visual features and constructs the Bridge Adapter; and the Vision Language CrossFormer fuses textual and visual features for RS-Scenes segmentation.

for remote sensing semantic segmentation. Many researchers have leveraged textual semantics to guide remote sensing scene segmentation [17]–[19]. However, there is a lack of thorough exploration regarding the effectiveness of the Caption [20], [21]. Meanwhile, due to the heterogeneity of remote sensing data, large language models (LLMs) have remained largely underexplored in this domain. FILM [22] utilizes textual semantics to guide multi-task image fusion, improving feature understanding and fusion effectiveness. RemoteCLIP [23] establishes a unified vision-language model for remote sensing, demonstrating its effectiveness in tasks such as image classification and image-text retrieval. SegCLIP [17] integrates mask classification with text-guided visual remote sensing semantic segmentation, enabling accurate identification of target object locations.

III. METHOD

A. Workflow Overview

Fig. 1 illustrates the overall pipeline of VMCSEg. The proposed framework consists of three main parts: (i) The MLLM generates remote sensing captions, and CLIP [24] is used to extract textual semantic information. (ii) DINOv3 [25] and a lightweight encoder extract visual features, while Bridge adapter is constructed to enable interaction between heterogeneous features. (iii) A Vision Language CrossFormer module is designed for fusing textual and visual semantics for dense remote sensing scene segmentation.

Our work fully leverages MLLM and VLM to obtain effective visual representations and corresponding textual semantic information, leading to a better understanding of relevant elements in remote sensing scenes. Finally, during the decoding stage, the fusion of intermediate and shallow visual representations results in improved segmentation performance.

B. MLLM Perceptual Understanding

A simple text prompt, such as a *photo of <class_name>*, is insufficient to accurately capture all the relationships between land-cover elements in RS-Scenes. Therefore, we employ a

Multimodal Large Language Model (MLLM) to generate more detailed and precise descriptions of the relevant land-cover elements in RS-scenes.

By designing appropriate *prompts*, the MLLM is used to generate textual captions describing the current scene. Simultaneously, the generated captions are encoded using the CLIP text encoder $\text{CLIP}_{\text{encoder_text}}$ to obtain semantic features.

C. Vision Feature Bridge Adapter

To effectively extract visual features from dense and overlapping remote sensing (RS) images $I \in \mathbb{R}^{3 \times H \times W}$, as shown in Fig. 1, we utilize DINOv3 to obtain general visual representations Φ_α . In addition, a lightweight encoder is constructed to capture scene-specific details and texture information, denoted as Φ_β . The formulas are as follows:

$$\Phi_\alpha = \text{DINOv3_Encoder}_\nu(I) \quad (1)$$

$$\Phi_\beta = \text{Lightweight_Encoder}_\nu(I) \quad (2)$$

To ensure that visual representations with different characteristics are fully communicated and integrated during training, and to make the VFM more suitable for RS tasks. As shown in Fig. 2, we design a Bridge Adapter, which performs fusion through the constructed Dilate Attention Adapter [26]. The formulas are as follows:

$$\Phi'_\alpha = \text{Dilate-Adapter}_\alpha(\Phi_\alpha) \quad (3)$$

$$\Phi'_\beta = \text{Dilate-Adapter}_\beta(\Phi_\beta) \quad (4)$$

$$\tilde{X} = \text{Conv}_{1 \times 1}(\text{Concat}(\Phi'_\alpha, \Phi'_\beta)) \quad (5)$$

where $\tilde{X}$ denotes the fused feature representation. In VMCSEg, a Bridge Adapter is employed across multiple levels, denoted as level_N , where $N \in 1, 2, 3, 4$, to optimally capture both shallow and deep visual features, facilitating enhanced feature decoding.

D. Vision Language CrossFormer

To enhance the interaction between visual representations and textual semantics, as depicted in Fig. 2, we propose a

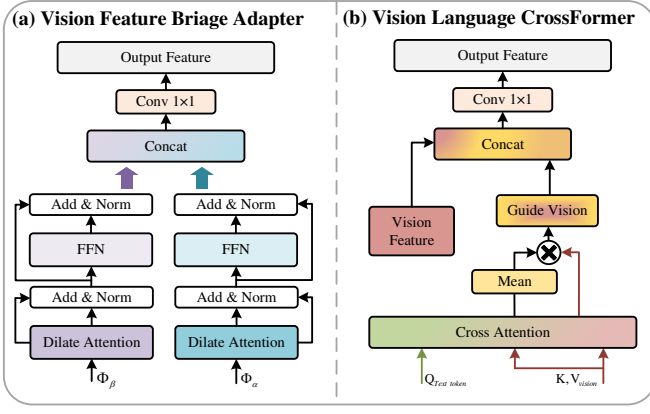


Fig. 2. Component details: (a) Structure of the Vision Feature Bridge Adapter; (b) Structure of the Vision Language CrossFormer.

Vision Language CrossFormer that enables textual information to more effectively guide the visual features attention towards the relevant segmentation target regions.

Visual features and textual semantics are initially projected to a common dimensionality. The visual features are employed to construct the keys and values, denoted as K, V_{vision} , which encompass the relevant attention regions, while the textual tokens serve as queries, denoted as $Q_{\text{Text token}}$. The corresponding formulas are as follows:

$$\tilde{F} = \text{Softmax}\left(\frac{Q_{\text{Text token}} K_{\text{vision}}^T}{\sqrt{d}}\right) V_{\text{vision}} \quad (6)$$

The $\tilde{F}$ values are mean-normalized to derive the corresponding weights, which are then used to guide the fusion of visual features $\tilde{X}$, resulting in the final feature representation.

However, relying solely on the textual guidance information $\tilde{F}$ without incorporating the original visual features $\tilde{X}$ is unreasonable. Therefore, we apply a 1×1 convolution to fuse the original features, resulting in visual feature $\tilde{L}$, which is then passed into the downstream segmentation decoder.

IV. EXPERIMENT

A. Dataset

Vaihingen dataset. The Vaihingen dataset [27] comprises 33 true orthophoto (TOP) images acquired using advanced airborne sensors, spanning an area of roughly 1.38 km² in Vaihingen. The images have a ground sampling distance (GSD) of approximately 9 cm, and each TOP image includes four channels: infrared, red, green, and blue (IRGB). Following [], we use 12 images for training and 4 images for testing. All images are cropped into 512×512 patches.

SynDrone dataset. The SynDrone dataset [28] contains urban drone imagery with both object-level and pixel-wise annotations. The dataset was acquired from drone flights over eight towns, selecting only images captured at a 90° viewing angle. Each image has a resolution of 1920×1080 and consists of three RGB channels. SynDrone provides fine-grained annotations to support detailed scene analysis. In this

work, we consider seven common land-cover classes and crop the images into 512×512 patches.

B. Implementation detail

Our model is implemented using the PyTorch framework. Visual features are extracted using a DINOv3 backbone with frozen parameters, initialized with LVD-0.3B distilled weights. For textual feature extraction, we employ the CLIP text encoder. We use the cross-entropy loss $\mathcal{L}_{ce}$ [29] for training. The model is optimized using the AdamW optimizer with an initial learning rate of 0.001, along with a multi-step learning rate scheduler. Scene-specific remote sensing captions are generated using the ChatGPT-4o MLLM. All experiments are conducted on an NVIDIA A800 GPU with 80 GB of memory.

C. Comparison with state-of-the-art

We evaluate the proposed method on two datasets with significantly different characteristics and conduct a comprehensive comparison with several state of the art semantic segmentation models, including MAREsUNet [10], UNetFormer [12], DC-Swin [30], A²-FPN [31], RS³Mamba [32], MetaSegNet [4], and SegCLIP [17].

Results on the Vaihingen Dataset: Table I presents the quantitative comparison on the Vaihingen dataset. The proposed method achieves superior overall performance, attaining an mIoU of 75.04% and an mF1 score of 85.33%, surpassing all competing approaches. Compared to the strongest baseline, RS³Mamba, our method improves the mIoU by 0.96% and the mF1 score by 0.62%, demonstrating the effectiveness of the proposed framework.

Notably, the proposed method shows clear advantages in challenging categories with complex appearance variations and small spatial extent. For the “Car” category, which is often confused with surrounding “impervious surface”, VMCSeg achieves 68.09% IoU and 81.02% F1 score, surpassing the second-best result of 65.34% IoU from A²FPN. Similarly, for Low vegetation, VMCSeg achieves 61.68% IoU, outperforming all baselines and improving the previous result 60.35%.

Results on the SynDrone Dataset: Table II shows the quantitative comparison on the dataset. The proposed method outperforms all others with an mIoU of 64.62% and an F1 of 72.82%. Compared to the strongest baseline, SegCLIP, our method improves mIoU by 2.24% and F1 by 1.35%, demonstrating the effectiveness of the proposed visual language fusion strategy.

More importantly, the proposed method demonstrates clear advantages in challenging small object categories. For the “person” class, characterized by small spatial extent and significant class imbalance, our method achieves 10.79% IoU and 19.49% F1, surpassing most existing methods while remaining competitive with the top baseline. Additionally, substantial improvements are observed in the vehicle and construction categories, with IoU scores of 69.92% and 86.59%, respectively, outperforming all compared approaches.

TABLE I
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE VAIHINGEN DATASET. * REPRESENTS TEXT MULTIMODAL METHODS.

Model	Backbone	Per-class IoU(%) / F1(%)					mIoU(%)	mF1(%)
		Impervious surfaces	Buildings	Low vegetation	Trees	Cars		
MAResUnet	ResNet-34	79.23/88.22	86.47/92.74	58.41/73.74	75.30/85.91	56.19/71.95	71.12	81.61
Unetformer	ResNet-18	76.19/86.49	84.63/91.67	55.63/71.49	74.42/85.33	51.68/68.14	68.51	80.63
A ² FPN	ResNet-18	79.59/88.64	88.50/93.90	59.30/74.45	75.77/86.22	65.34/79.04	73.82	84.45
RS ³ Mamba	VMamba-T	79.87/88.81	88.76/94.05	<u>60.35/75.27</u>	<u>76.56/86.72</u>	<u>64.85/78.68</u>	<u>74.08</u>	<u>84.71</u>
DC-Swin	Swin-B	72.78/84.24	81.72/89.94	52.86/69.16	73.05/84.42	48.69/65.49	65.82	77.35
MetaSegNet*	Swin-B	73.41/84.67	82.03/90.13	55.96/71.76	73.03/84.41	47.49/64.39	66.38	79.07
SegCLIP*	Swin-B	76.88/86.93	85.17/91.99	59.20/74.37	76.19/86.49	63.71/77.84	72.23	81.27
VMCSeg*(Ours)	DINOv3-LVD	79.40/88.52	<u>88.57/93.94</u>	61.68/76.30	77.47/87.30	68.09/81.02	75.04	85.33

TABLE II
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE SYNDROME DATASET. * REPRESENTS TEXT MULTIMODAL METHODS.

Model	Backbone	Per-class IoU(%) / F1(%)							mIoU(%)	mF1(%)
		road	nature	person	vehicle	construction	obstacle	water		
MAResUnet	ResNet-34	80.29/89.07	88.67/93.99	1.34/4.68	39.98/57.12	57.26/72.82	9.63/17.57	77.87/87.56	50.72	60.4
Unetformer	ResNet-18	77.10/87.07	82.50/90.41	0.40/0.80	22.03/36.11	45.28/62.33	9.66/17.61	73.47/84.71	44.35	54.15
A ² FPN	ResNet-18	82.51/90.42	<u>89.83/94.64</u>	10.25/18.60	59.98/74.99	61.67/76.29	23.00/37.40	<u>81.33/89.70</u>	58.37	68.86
RS ³ Mamba	VMamba-T	82.91/90.66	89.29/94.34	2.83/5.49	53.21/69.46	53.21/69.46	16.29/28.02	81.26/89.66	55.17	64.71
DC-Swin	Swin-B	80.24/89.04	88.29/93.78	0.16/0.31	27.89/43.62	54.77/70.78	9.72/17.71	74.17/85.17	47.89	57.2
MetaSegNet*	Swin-B	84.36/91.52	91.90/95.78	1.54/4.82	45.26/62.31	72.51/84.07	4.62/8.83	73.36/84.63	53.36	61.71
SegCLIP*	Swin-B	89.82/94.64	89.82/94.64	12.56/22.31	61.85/76.43	<u>74.90/85.65</u>	19.97/33.29	83.84/91.21	62.38	71.47
VMCSeg*(Ours)	DINOv3-LVD	<u>89.31/94.36</u>	93.81/96.81	<u>10.79/19.49</u>	69.92/82.30	86.59/92.81	<u>20.75/34.37</u>	81.14/89.59	64.62	72.82

D. Main Components Analysis

1) *The performance of Briage-Adapter*: Although DINOv3 primarily extracts generic dense representations, it fails to fully capture fine-grained boundary details and texture information in complex remote sensing scenes. Consequently, the design of a lightweight detail encoder becomes necessary. In addition, the Bridge-Adapter enables the interactive fusion of heterogeneous land cover representations, allowing DINOv3 to more effectively support downstream segmentation tasks. As shown in Table I, our method achieves strong segmentation performance not only for large area categories such as “Building,” but also attains the best results for densely distributed classes such as “Tree.”

2) *Caption Generation and Extraction*: Unlike text generated from fixed templates, captions produced by a Multimodal Large Language Model better preserve the spatial relationships among different land cover categories in remote sensing scenes. The resulting RS caption provides interpretable textual descriptions of the land cover present in the current scene. We encode the caption using CLIP to obtain effective textual semantic representations, which are used to guide the extraction of visual features. As shown in Table I, our method achieves superior segmentation performance on small scale and low sample categories such as “Car.”

3) *Vision Language CrosFormer*: In this work, we design Vision Language CrosFormer with a concise architectural framework that effectively leverages the perceptual advantages of textual semantics to guide the segmentation of extracted fine grained visual features. During training, we construct multi

layer cross modal attention mechanisms to fully exploit the scene level information conveyed by textual semantics. As shown in Table II, our method achieves favorable segmentation performance even on challenging categories such as “person” and “obstacle,” demonstrating the superiority of the RS Caption guided strategy.

V. CONCLUSION

To overcome the limitations of single modality learning in complex remote sensing scene segmentation, we propose VMCSeg, which leverages a Multimodal Large Language Model to perceive and interpret complex remote sensing scenes and generate scene aware RS captions. To address the shortcoming that the visual foundation model primarily extracts generic feature representations, we introduce a lightweight encoder to capture fine grained texture information. In addition, a Bridge Adapter is designed to facilitate the interactive fusion of heterogeneous features, making DINOv3 more suitable for remote sensing tasks. Furthermore, we construct a Vision Language CrossFormer to integrate textual semantics with visual information, enabling visual features to focus more effectively on category specific characteristics that benefit downstream segmentation tasks. During the decoding stage, we exploit not only deep features enhanced by textual semantics but also intermediate and shallow visual features for segmentation. Extensive experiments demonstrate that the proposed model achieves superior segmentation performance on the public Vaihingen and SynDrone datasets.

REFERENCES

- [1] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo *et al.*, “Segment anything,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2023, pp. 4015–4026.
- [2] J. Long, E. Shelhamer, and T. Darrell, “Fully convolutional networks for semantic segmentation,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2015, pp. 3431–3440.
- [3] O. Ronneberger, P. Fischer, and T. Brox, “U-net: Convolutional networks for biomedical image segmentation,” in *Medical image computing and computer-assisted intervention—MICCAI 2015: 18th international conference, Munich, Germany, October 5–9, 2015, proceedings, part III 18*, 2015, pp. 234–241.
- [4] L. Wang, S. Dong, Y. Chen, X. Meng, S. Fang, and S. Fei, “Metasegnet: Metadata-collaborative vision-language representation learning for semantic segmentation of remote sensing images,” *IEEE Transactions on Geoscience and Remote Sensing*, 2024.
- [5] M. Zhu, Y. Tian, H. Chen, C. Zhou, Q. Guo, Y. Liu, M. Yang, and C. Shen, “Segagent: Exploring pixel understanding capabilities in mllms by imitating human annotator trajectories,” in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 3686–3696.
- [6] X. Li, B. Zhong, Q. Liang, Z. Mo, J. Nong, and S. Song, “Dynamic updates for language adaptation in visual-language tracking,” in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 19 165–19 174.
- [7] Q. Zhu, J. Lao, D. Ji, J. Luo, K. Wu, Y. Zhang, L. Ru, J. Wang, J. Chen, M. Yang *et al.*, “Skysense-o: Towards open-world remote sensing interpretation with vision-centric visual-language modeling,” in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 14 733–14 744.
- [8] H. Li, K. Qiu, L. Chen, X. Mei, L. Hong, and C. Tao, “Scattnet: Semantic segmentation network with spatial and channel attention mechanism for high-resolution remote sensing images,” *IEEE Geoscience and Remote Sensing Letters*, vol. 18, no. 5, pp. 905–909, 2020.
- [9] L. Wang, C. Zhang, R. Li, C. Duan, X. Meng, and P. M. Atkinson, “Scale-aware neural network for semantic segmentation of multi-resolution remote sensing images,” *Remote sensing*, vol. 13, no. 24, p. 5015, 2021.
- [10] R. Li, S. Zheng, C. Duan, J. Su, and C. Zhang, “Multistage attention resu-net for semantic segmentation of fine-resolution remote sensing images,” *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2021.
- [11] R. Li, S. Zheng, C. Zhang, C. Duan, L. Wang, and P. M. Atkinson, “Abcnet: Attentive bilateral contextual network for efficient semantic segmentation of fine-resolution remotely sensed imagery,” *ISPRS journal of photogrammetry and remote sensing*, vol. 181, pp. 84–98, 2021.
- [12] L. Wang, R. Li, C. Zhang, S. Fang, C. Duan, X. Meng, and P. M. Atkinson, “Unetformer: A unet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery,” *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 190, pp. 196–214, 2022.
- [13] P. Song, J. Li, Z. An, H. Fan, and L. Fan, “Ctmfnet: Cnn and transformer multiscale fusion network of remote sensing urban scene imagery,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1–14, 2022.
- [14] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, “Swin transformer embedding unet for remote sensing image semantic segmentation,” *IEEE transactions on geoscience and remote sensing*, vol. 60, pp. 1–15, 2022.
- [15] X. Ma, Q. Wu, X. Zhao, X. Zhang, M.-O. Pun, and B. Huang, “Sam-assisted remote sensing imagery semantic segmentation with object and boundary constraints,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–16, 2024.
- [16] X. Ma, X. Zhang, M.-O. Pun, and B. Huang, “A unified framework with multimodal fine-tuning for remote sensing semantic segmentation,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 63, pp. 1–15, 2025.
- [17] S. Zhang, B. Zhang, Y. Wu, H. Zhou, J. Jiang, and J. Ma, “Segclip: Multimodal visual-language and prompt learning for high-resolution remote sensing semantic segmentation,” *IEEE Transactions on Geoscience and Remote Sensing*, 2024.
- [18] J. Lu, D. Batra, D. Parikh, and S. Lee, “Vilbert: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks,” *Advances in neural information processing systems*, vol. 32, 2019.
- [19] J. Li, R. Selvaraju, A. Gotmare, S. Joty, C. Xiong, and S. C. H. Hoi, “Align before fuse: Vision and language representation learning with momentum distillation,” *Advances in neural information processing systems*, vol. 34, pp. 9694–9705, 2021.
- [20] R. Zhao, Z. Shi, and Z. Zou, “High-resolution remote sensing image captioning based on structured attention,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–14, 2021.
- [21] X. Ye, S. Wang, Y. Gu, J. Wang, R. Wang, B. Hou, F. Giunchiglia, and L. Jiao, “A joint-training two-stage method for remote sensing image captioning,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–16, 2022.
- [22] Z. Zhao, L. Deng, H. Bai, Y. Cui, Z. Zhang, Y. Zhang, H. Qin, D. Chen, J. Zhang, P. Wang *et al.*, “Image fusion via vision-language model,” *arXiv preprint arXiv:2402.02235*, 2024.
- [23] F. Liu, D. Chen, Z. Guan, X. Zhou, J. Zhu, Q. Ye, L. Fu, and J. Zhou, “Remoteclip: A vision language foundation model for remote sensing,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–16, 2024.
- [24] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, “Learning transferable visual models from natural language supervision,” in *International conference on machine learning*. PmLR, 2021, pp. 8748–8763.
- [25] O. Siméoni, H. V. Vo, M. Seitzer, F. Baldassarre, M. Oquab, C. Jose, V. Khalidov, M. Szafraniec, S. Yi, M. Ramamonjisoa *et al.*, “Dinov3,” *arXiv preprint arXiv:2508.10104*, 2025.
- [26] J. Jiao, Y.-M. Tang, K.-Y. Lin, Y. Gao, A. J. Ma, Y. Wang, and W.-S. Zheng, “Dilateformer: Multi-scale dilated transformer for visual recognition,” *IEEE transactions on multimedia*, vol. 25, pp. 8906–8919, 2023.
- [27] S. Waqas Zamir, A. Arora, A. Gupta, S. Khan, G. Sun, F. Shahbaz Khan, F. Zhu, L. Shao, G.-S. Xia, and X. Bai, “isaid: A large-scale dataset for instance segmentation in aerial images,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2019, pp. 28–37.
- [28] G. Rizzoli, F. Barbato, M. Caligiuri, and P. Zanuttigh, “Syndrone-multi-modal uav dataset for urban scenarios,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 2210–2220.
- [29] A. Mao, M. Mohri, and Y. Zhong, “Cross-entropy loss functions: Theoretical analysis and applications,” in *International conference on Machine learning*, 2023, pp. 23 803–23 828.
- [30] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, “A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images,” *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2022.
- [31] M. Hu, Y. Li, L. Fang, and S. Wang, “A2-fpn: Attention aggregation based feature pyramid network for instance segmentation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 15 343–15 352.
- [32] X. Ma, X. Zhang, and M.-O. Pun, “Rs 3 mamba: Visual state space model for remote sensing image semantic segmentation,” *IEEE Geoscience and Remote Sensing Letters*, vol. 21, pp. 1–5, 2024.